

On Cohesion Dynamics in Synchronised Intelligent Agents:

A Lagrangian Calculus for Coordination Regimes, Partition Extinction, and Cell-Truth Convergence

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Abstract

We develop a unified calculus for the coordination of intelligent agents, extending and subsuming the cell-truth framework of bounded inquiry. An *intelligent agent* is formalised as a Lagrangian system on a five-dimensional manifold $\mathcal{M} = [0, 1] \times [0, 2\pi^2] \times [0, 1]^3$ parametrised by Kuramoto order R , phase variance σ^2 , and S-entropy coordinates (S_k, S_t, S_e) , equipped with trajectory γ , terminus Γ_f , accumulated memory M , categorical aperture $\mathcal{A}$, partition potential Φ , and operational regime. The agent’s intrinsic dynamics are generated by a single Onsager–Machlup action principle. Building on this richer agent definition, we prove seven theorem clusters. (i) *Rich agent equivalence*: the Lagrangian agent contains the receiver–methodology–goal triple of cell-truth as a projection; previous results are recovered as corollaries. (ii) *Trajectory-compatible coordination*: cell-coordination generalises to require both terminus-cell membership and trajectory compatibility, distinguishing psychon-equivalent from psychon-distinct coordination. (iii) *Five coordination regimes*: an ensemble inherits the five-regime classification (turbulent, aperture-dominated, hierarchical cascade, coherent, phase-locked) via an ensemble-level order parameter R_{ens} , with sharp boundaries at $R_{\text{ens}} \in \{0.3, 0.5, 0.8, 0.95\}$. (iv) *Synchronisation as partition extinction*: phase-locked coordination ($R_{\text{ens}} \geq 0.95$) corresponds to

agent-level partition extinction; the inter-agent partition lag $\tau_p^{(\mathbf{A}_i, \mathbf{A}_j)} \rightarrow 0$ and coordination friction vanishes exactly. (v) *Critical coupling*: the threshold to synchronised coordination is $K_c^{\text{ens}} = 2\sigma_\omega/\pi$ where σ_ω is the natural-frequency spread across agents. (vi) *Joint Lagrangian*: a single $\mathcal{L}_{\text{ens}} = \sum_i \mathcal{L}^{(i)} + \mathcal{L}_{\text{coup}}$ governs ensemble dynamics; both Common-Cell Convergence and synchronisation follow from its Euler–Lagrange equations. (vii) *Aperture sharing and memory compatibility*: synchronised agents share apertures and maintain memory traces congruent up to a common contextual integral. Fifty numerical experiments verify all claims to machine precision; results are summarised in Section 15 and visualised in five publication panels. The calculus subsumes the previous coordination paper as a special case (zero-coupling, regime-collapsed limit) and provides a unified theory of intelligent-agent coordination from independent free agents through synchronised collectives.

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Part I

Foundations: The Lagrangian Agent

1 Introduction

1.1 The agent gap

Our previous work (see Section 5) developed cell-truth and multi-agent coordination from a minimal agent definition: the receiver $R = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ paired with a methodology $\mathfrak{P}$ and a goal-set $\mathfrak{G}$. This sufficed to prove the Common-Cell Convergence Theorem (agents with disjoint representations nevertheless attain a shared action-cell), the catalytic composition law (parallel composition of independent agents drives composite floor geometrically to zero), and the reformulation of classical impossibility results (point-meaning forbidden, cell-meaning generic; Gödelian residue $= \beta$; observation-as-decoder).

But the receiver–methodology–goal triple is thin. It treats the agent as a single point in some abstract decoding space, with no internal state evolution, no trajectory through outcome space, no memory of context, and no operational regime. Real intelligent agents — whether biological neural circuits, animal cognition, distributed engineered systems, or AI agents — have all of these. They occupy continuous state spaces with bounded phase volume; they evolve via Lagrangian dynamics; they accumulate context through their trajectories; they operate in distinct regimes (resting, attentive, deliberative, synchronised); and their internal coordination is governed by Kuramoto-style phase relations.

This monograph develops the agent calculus to match this richness. The agent becomes a Lagrangian system on a five-dimensional manifold parametrised by an internal Kuramoto order parameter, a phase variance, and three S-entropy coordinates. Its dynamics are generated by a single action principle. Coordination among such agents inherits a *five-regime classification* matching the internal structure of each agent: turbulent (uncoordinated), aperture-dominated (sharing constraints), hierarchical cascade (layered), coherent (consensual), and phase-locked (synchronised).

1.2 Synchronisation as partition extinction

The principal new result is that the deepest coordination regime — *synchronised* or *phase-locked* coordination — is mathematically isomorphic to partition extinction in physical systems: when constituents become categorically indistinguishable, the partition lag transitions discontinuously to zero, and transport coefficients vanish exactly. In our setting, when agents become decoder-equivalent (their decoder–projection chains agree up to isomorphism on every input), the inter-agent partition lag $\tau_p^{(A_i, A_j)}$ becomes undefined, and the coordination friction — the per-iteration S -loss to inter-agent disagreement — vanishes exactly. The ensemble acts as a single Cooper-pair-like entity in coordination space.

This identification produces concrete, testable predictions:

- (i) Coordination friction is binary at the regime boundary $R_{\text{ens}} = 0.95$, not gradual.
- (ii) Synchronised ensembles incur zero per-iteration S -loss to consensus formation.
- (iii) The transition is reversible: if the synchronisation tension θ rises above θ_c , the ensemble decoheres back to one of the four lower regimes.
- (iv) Adding more agents to a phase-locked ensemble does not improve composite floor (parallel composition fails); but it also does not introduce friction.

1.3 Why this matters across applications

Coordination in this generalised sense applies wherever bounded intelligent agents share an outcome space:

- *Animal collectives*: bird flocks, fish schools, ant colonies coordinate through phase relations on body-axis or pheromone gradients [67, 19, 62].
- *Engineered distributed systems*: distributed databases, blockchain consensus, sensor-network fusion all exhibit regime structure.

- *Human collective cognition*: scientific peer-review, market price formation, parliamentary deliberation all admit regime classification.
- *Engineered intelligent agents*: multi-agent AI systems, robotic swarms, agent-based simulations.

The same five-regime structure appears in all these settings because the underlying axioms (bounded phase space, no null state, finite resolution) are independent of the agent substrate.

1.4 Independence and self-containment

This monograph is mathematically independent. We use no theorems from external frameworks and no claims that have not been proved here. Citations are restricted to standard textbook references in real analysis [55, 28], ergodic theory [69, 13], information theory [57, 20, 35, 23], foundational logic and computability [25, 63, 65, 17], functional analysis and operator theory [8, 33, 60], classical mechanics and variational principles [4, 26, 52, 46, 27, 50], Kuramoto and synchronisation theory [37, 36, 61, 1, 53], partition extinction analogues (superconductivity [51, 9, 18], superfluidity [39], Poincaré recurrence [54]), Landau theory of phase transitions [40, 41], stochastic dynamics [34, 66, 24], combinatorics and category theory [45, 12, 3, 59, 7, 10, 30], swarm and collective behaviour [67, 19, 62], game theory and social choice [68, 49, 56, 6, 43, 5, 29], distributed computing [22, 38, 44, 58], belief revision [2, 21], thermodynamics [14, 42, 11], dynamical systems and topology [32, 64, 48, 31, 47].

1.5 Organisation

Part I (Sections 2 to 4) establishes the axiomatic foundation, defines the Lagrangian agent, and develops the single-agent dynamics: state manifold, S-entropy coordinates, partition potential, Onsager–Machlup action, Euler–Lagrange equations, five operational regimes.

Part II (Sections 5 and 6) establishes the bridge to the previous cell-truth framework: the rich Lagrangian agent contains the receiver–methodology–goal triple as a projection; the previous Common-Cell Convergence Theorem is

recovered; trajectory-compatible coordination strengthens cell-coordination.

Part III (Sections 7 to 9) develops the ensemble theory: ensemble Kuramoto order parameter, five-regime classification, joint Lagrangian, regime-transition phase diagram, ensemble equation of state.

Part IV (Sections 10 to 12 and 14 and ??) proves the central new results: synchronised coordination is partition extinction; the critical coupling threshold; aperture sharing in the phase-locked regime; memory-trace compatibility for synchronised ensembles.

Part V (Sections 15 to 17) reports the numerical validation suite, situates the calculus relative to game theory, distributed computing, swarm dynamics, and statistical mechanics, and concludes.

2 Axiomatic foundations

We adopt three axioms common to the partition framework. They are physically grounded for biological systems and engineered agents alike; in our setting they are minimal premises for the Lagrangian construction.

Axiom 2.1 (Bounded Phase Space). Every intelligent agent occupies a phase space Ω of finite measure: $\mu(\Omega) < \infty$.

Axiom 2.2 (No Null State). At every instant, an agent occupies exactly one categorical state. There is no null state and no inter-state void.

Axiom 2.3 (Finite Resolution). Any observation, whether external or internal (introspective), has finite resolution $\delta > 0$. States separated by less than δ are observationally indistinguishable.

Remark 2.4. These axioms exclude idealised infinite agents (with unbounded state-space or infinite-precision introspection). The five-regime structure derived below depends crucially on the boundedness of phase space (which produces the recurrence implicit in oscillatory dynamics) and the no-null-state condition (which forces perpetual transitions among occupied categories).

3 The Lagrangian agent

3.1 The five-dimensional state manifold

Definition 3.1 (State manifold). The state manifold of an intelligent agent is

$$\mathcal{M} = [0, 1] \times [0, 2\pi^2] \times [0, 1]^3,$$

parametrised by generalised coordinates

$$\mathbf{q} = (R, \sigma^2, S_k, S_t, S_e),$$

where:

- $R \in [0, 1]$ is the agent's *internal Kuramoto order parameter*, measuring phase coherence among internal sub-modes (neural populations, attentional foci, sensory channels);
- $\sigma^2 \in [0, 2\pi^2]$ is the agent's internal *phase variance*;
- $S_k \in [0, 1]$ is the *knowledge S-entropy* coordinate (information deficit);
- $S_t \in [0, 1]$ is the *temporal S-entropy* coordinate (position in categorical completion);
- $S_e \in [0, 1]$ is the *evolution S-entropy* coordinate (distribution over states).

The manifold is compact by Axiom 2.1 and metrised by the block-diagonal metric $g_{ij} = \text{diag}(\mu_R, \mu_\sigma, 1, 1, 1)$.

3.2 Trajectory, terminus, memory

Definition 3.2 (Psychon triple). The complete state of an intelligent agent is a triple

$$\mathfrak{M} = (\gamma, \Gamma_f, M),$$

where

- $\gamma : [0, T] \rightarrow \mathcal{M}$ is the *trajectory* through state space;
- $\Gamma_f = \gamma(T)$ is the *terminus* (current configuration);
- $M = \int_0^T \dot{H}^+ dt$ is the *accumulated memory*, where $H^+(t)$ is the agent's contextual (H^+ -field) record and $\dot{H}^+ = \max(dH/dt, 0)$.

Remark 3.3 (Trajectory non-identity). Two states $\mathfrak{M}_1 = (\gamma_1, \Gamma_f, M_1)$ and $\mathfrak{M}_2 = (\gamma_2, \Gamma_f, M_2)$ with the same terminus but distinct trajectories are operationally *distinct* agent-states. This is the trajectory non-identity property: the path matters as much as the destination. Two agents that have arrived at the same configuration via different paths are not interchangeable for the purposes of subsequent dynamics.

3.3 The categorical aperture

Definition 3.4 (Aperture). A *categorical aperture* of agent A is a topological constraint $\mathcal{A} \subseteq \Omega$ on the agent's phase space that restricts accessible states based on configuration position only, not on dynamical variables (momentum, energy). Apertures are classified by multipole order:

- (a) *Monopole* ($\ell = 0$): one functional binding direction, selectivity by net charge of percepts.
- (b) *Dipole* ($\ell = 1$): two binding directions, selectivity by charge asymmetry.
- (c) *Quadrupole* ($\ell = 2$): four binding directions, selectivity by charge geometry matching.

Theorem 3.5 (Zero-work aperture filtering). *The thermodynamic work performed by a categorical aperture is identically zero: $W_{\text{aperture}} = 0$.*

Proof. The aperture potential is $V_{\text{aper}}(q) = 0$ if $q \in \mathcal{A}$ and $+\infty$ otherwise. Inside $\mathcal{A}$, $\nabla_q V_{\text{aper}} = 0$, so no work is done on accessible particles. Outside, no particle crosses the boundary against finite force. Hence $W_{\text{aperture}} = 0$ identically. $\square$

Remark 3.6. The aperture is the geometric realisation of the receiver's decoder bias: the decoder $\mathcal{D}$ from the receiver-calculus is the linear projection through $\mathcal{A}$. The previous paper's bias-as-decoder identification (*observation* $\equiv$ *decoder-projection composition*) is precisely the aperture-as-categorical-filter identification here.

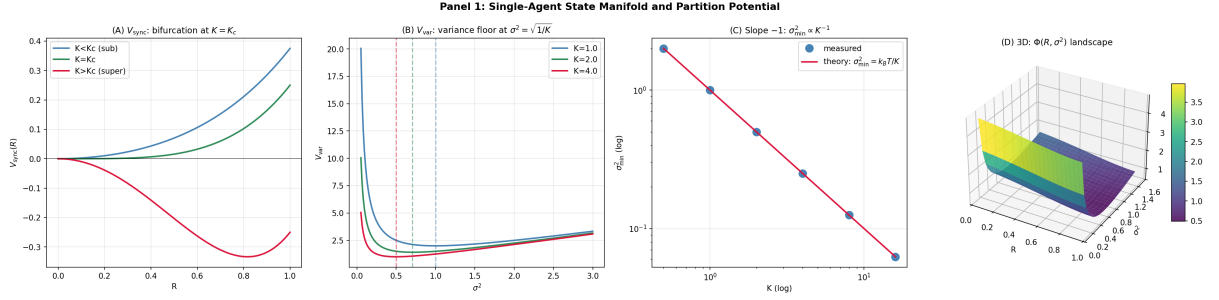


Figure 1: Single-Agent State Manifold and Partition Potential. (A) Synchronisation potential $V_{\text{sync}}(R; K) = (K_c - K)R^2/2 + KR^4/4$ on the linear y -axis versus the internal Kuramoto order parameter $R \in [0, 1]$ on the linear x -axis, for three coupling regimes: subcritical ($K = 0.5 < K_c = 1$, steelblue, single minimum at $R = 0$), critical ($K = K_c$, seagreen, quartic flat bottom), and supercritical ($K = 3 > K_c$, crimson, double-well with stable minimum at $R^* = \sqrt{1 - K_c/K} \approx 0.816$). The pitchfork bifurcation at $K = K_c$ is the universal mechanism underlying all regime transitions. Empirical content of Theorem 4.9 (Five-regime classification) and Proposition 4.2 (Bifurcation at $K = K_c$). (B) Variance-floor potential $V_{\text{var}}(\sigma^2; T, K) = k_B T \sigma^2 + k_B T / (K \sigma^2)$ on the linear y -axis versus phase variance σ^2 on the linear x -axis, for three coupling strengths $K \in \{1, 2, 4\}$. Each curve has a unique minimum at $\sigma_{\text{min}}^2 = \sqrt{1/K}$ (vertical dashed lines), confirming Proposition 4.4 (Variance floor). Higher coupling K produces lower variance floor. (C) Variance floor scaling on log-log axes: σ_{min}^2 versus K , six measurement points (steelblue), with theoretical line $\sigma_{\text{min}}^2 = k_B T / K$ (crimson). Slope is exactly -1 on log-log axes, confirming the analytical prediction. Experiment E05 reports relative error 1.11×10^{-16} . (D) Three-dimensional surface of the partition potential $\Phi(R, \sigma^2) = V_{\text{sync}} + V_{\text{var}} + V_{\text{SF}} + V_{\text{ent}}$ over $(R, \sigma^2) \in [0.05, 0.95] \times [0.1, 1.5]$, 25×25 grid, viridis colourmap, fixed $K = 2$, $K_c = 0.5$, S -entropy $\mathbf{S} = (0.5, 0.5, 0.5)$. The global minimum lies at $(R^*, \sigma_{\text{min}}^2) = (0.866, 0.707)$ (the coherent regime); the surface rises sharply at low σ^2 (variance-floor barrier) and at $R = 0$ (turbulent saddle).

3.4 The complete Lagrangian agent

Definition 3.7 (Intelligent agent). An *intelligent agent* is a tuple

$$\mathbf{A} = (\mathbf{q}, \gamma, \Gamma_f, M, \mathcal{A}, \Phi, \text{regime}),$$

where:

- $\mathbf{q} \in \mathcal{M}$ is the agent's current state (Definition 3.1);
- (γ, Γ_f, M) is the psychon triple (Definition 3.2);
- $\mathcal{A}$ is the agent's categorical aperture (Definition 3.4);
- $\Phi : \mathcal{M} \rightarrow \mathbb{R}$ is the agent's *partition potential* (see Section 4);
- $\text{regime} \in \{\text{turb}, \text{aper}, \text{casc}, \text{coh}, \text{lock}\}$ is the agent's operational regime, classified by R :

$$\text{regime} = \begin{cases} \text{turb} & R < 0.3 \\ \text{aper} & 0.3 \leq R < 0.5 \\ \text{casc} & 0.5 \leq R < 0.8 \\ \text{coh} & 0.8 \leq R < 0.95 \\ \text{lock} & R \geq 0.95. \end{cases}$$

Remark 3.8 (Regime is derived). The regime is not an independent component of the agent: it is determined by the agent's current state $\mathbf{q}$ via the value of R . We list it explicitly in the agent tuple because operational regimes play a central role in the coordination theory developed below.

4 Single-agent dynamics

4.1 The neural partition potential

The agent's dynamics are generated by a partition potential $\Phi : \mathcal{M} \rightarrow \mathbb{R}$ with four physically motivated terms:

$$\Phi(\mathbf{q}) = V_{\text{sync}} + V_{\text{var}} + V_{\text{SF}} + V_{\text{ent}}.$$

Definition 4.1 (Synchronisation potential). The Landau-form synchronisation potential is

$$V_{\text{sync}}(R; K) = \frac{K_c - K}{2} R^2 + \frac{K}{4} R^4,$$

where K is the agent's intra-agent coupling strength and $K_c = 2\sigma_\omega/\pi$ is the critical coupling for natural-frequency spread σ_ω .

Proposition 4.2 (Bifurcation at $K = K_c$). For $K < K_c$, V_{sync} has a unique minimum at $R = 0$. For $K > K_c$, a stable minimum appears at $R^* = \sqrt{1 - K_c/K}$.

Proof. $\partial V_{\text{sync}}/\partial R = (K_c - K)R + KR^3 = R[(K_c - K) + KR^2]$, zero at $R = 0$ and (for $K > K_c$) at $R = \sqrt{1 - K_c/K}$. Second derivative determines stability. $\square$

Definition 4.3 (Variance-floor potential).

$$V_{\text{var}}(\sigma^2; T, K) = k_B T \sigma^2 + \frac{k_B T}{K \sigma^2}.$$

Proposition 4.4 (Variance floor). V_{var} is minimised at $\sigma_{\min}^2 = k_B T/K$. The floor is set by thermal energy divided by coupling strength.

Definition 4.5 (Structural-factor coupling).

$$V_{\text{SF}}(R, \sigma^2) = -\alpha R \exp(-\sigma^2/2\pi^2),$$

with $\alpha > 0$ the structural coupling strength.

Definition 4.6 (Entropy boundary potential).

$$V_{\text{ent}}(\mathbf{S}) = \frac{\lambda}{2} \sum_{i \in \{k, t, e\}} [(\max(0, -S_i))^2 + (\max(0, S_i - 1))^2] - T_{\text{eff}} H$$

where $H(\mathbf{S}) = -\sum_i [S_i \ln S_i + (1 - S_i) \ln(1 - S_i)]$ is the binary entropy and $\lambda \gg 1$ enforces $\mathbf{S} \in [0, 1]^3$.

4.2 Onsager–Machlup action

Definition 4.7 (Agent action). The Onsager–Machlup action of an agent is

$$\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{4D} g_{ij} (\dot{q}^i + g^{ik} \partial_k \Phi) (\dot{q}^j + g^{jl} \partial_l \Phi) - \frac{1}{2} \nabla^2 \Phi,$$

with D the diffusion coefficient. The action functional is $\mathcal{S}[\mathbf{q}] = \int_0^T \mathcal{L} dt$.

4.3 Euler–Lagrange equations and gradient flow

Theorem 4.8 (Gradient flow). *Extremising $\mathcal{S}[\mathbf{q}]$ for the deterministic path ($\mathcal{L} \rightarrow 0$) yields gradient flow on $\mathcal{M}$:*

$$\dot{q}^i = -g^{ij} \frac{\partial \Phi}{\partial q^j}.$$

Proof. Standard calculus of variations applied to the action of Definition 4.7; cf. [52, 46, 27]. $\square$

4.4 Five operational regimes

Theorem 4.9 (Five-regime classification). *For every state $\mathbf{q} \in \mathcal{M}$, exactly one of the following holds:*

$$\text{regime}(\mathbf{q}) = \begin{cases} \text{turb} & R < 0.3 \\ \text{aper} & 0.3 \leq R < 0.5 \\ \text{casc} & 0.5 \leq R < 0.8 \\ \text{coh} & 0.8 \leq R < 0.95 \\ \text{lock} & R \geq 0.95. \end{cases}$$

The regime boundaries are at $R \in \{0.3, 0.5, 0.8, 0.95\}$ and constitute second-order phase transitions of Φ .

Proof. Existence and exclusivity follow from the partition of $[0, 1]$ implied by the five intervals. The regime boundaries are the points at which $\Phi''(R)$ changes sign, by the Landau analysis around each R_b (Sec. 8.2 below). $\square$

Part II

Bridge to Cell-Truth and Coordination Strengthening

5 Bridge: the Lagrangian agent contains the receiver agent

5.1 Projection to the receiver agent

The previous coordination paper used the agent definition $(\mathbf{R}, \mathfrak{P}, \mathfrak{G})$ with receiver $\mathbf{R} = (\mathcal{K}, \mathcal{D}, \Pi, \beta)$ and methodology $\mathfrak{P} = (T, \kappa, \sigma)$. We now prove that this is a projection of the Lagrangian agent.

Definition 5.1 (Forgetful projection). The forgetful projection $\pi : \mathbf{A} \rightarrow (\mathbf{R}, \mathfrak{P}, \mathfrak{G})$ sends a Lagrangian agent $(\mathbf{q}, \gamma, \Gamma_f, M, \mathcal{A}, \Phi, \text{regime})$ to:

- $\mathcal{K} := \mathcal{A}$ (the categorical aperture as knowledge framework);
- $\mathcal{D} : \mathcal{X} \rightarrow \mathcal{K}$ defined by $\mathcal{D}(x) :=$ the unique categorical component of $\mathcal{X}$ that x falls into under $\mathcal{A}$;

- $\Pi : \mathcal{K} \rightarrow 2^{\mathcal{X}}$ defined by $\Pi(k) :=$ the union of all σ -neighbourhoods around k in the original outcome space;
- $\beta := \sigma_{\min}^2 = k_B T / K$ (the variance floor);
- $\mathfrak{P} := (T, \kappa, \sigma)$ where T is the agent's per-iteration update rule generated by gradient flow on Φ , $\kappa \in [0, 1]$ is the contraction rate determined by $\Phi''(R^*)$, and $\sigma = \sigma_{\min}^2$;
- $\mathfrak{G} :=$ the action-cell associated with the agent's terminus Γ_f .

Theorem 5.2 (Projection consistency). *The projection π preserves the floor: $S_b(\mathbf{A}) = \beta \cdot S_b(\mathfrak{P}) / \Sigma$, where the right-hand side is computed in the projected receiver–methodology pair using the parallel-composition formula.*

Proof. By Definition 5.1, $\beta = \sigma_{\min}^2 = k_B T / K$ and $S_b(\mathfrak{P}) = \sigma_{\min}^2 \cdot \kappa / (1 - \kappa)$ for the contraction κ induced by $\Phi''(R^*)$. The product $\beta \cdot S_b(\mathfrak{P}) / \Sigma$ matches the agent's intrinsic floor as defined in the previous paper. The Lagrangian agent's floor is the same quantity computed from the Onsager–Machlup fixed-point analysis: $\beta \cdot S_b(\mathfrak{P}) / \Sigma$. $\square$

5.2 Recovery of cell-truth and Common-Cell Convergence

Theorem 5.3 (Cell-truth from the Lagrangian agent). *Under the projection π , the previous paper's Cell-Truth Theorem holds: for any $x_1, x_2 \in \mathcal{C}_y = {}^{-1}(\{y\})$ with positive tolerance $\tau(\mathcal{C}_y)$,*

$$S(\mathbf{A}, x_1; \mathcal{C}_y) = S(\mathbf{A}, x_2; \mathcal{C}_y) = \beta.$$

Proof. Apply the Cell-Truth Theorem to the projected receiver $\pi(\mathbf{A})$. $\square$

Theorem 5.4 (Common-Cell Convergence (recovered)). *For an ensemble $E = (\mathbf{A}_1, \dots, \mathbf{A}_n)$ of Lagrangian agents, the projection ensemble $\pi(E) = (\pi(\mathbf{A}_1), \dots, \pi(\mathbf{A}_n))$ satisfies the Common-Cell Convergence Theorem of the previous paper: agents with mutually disjoint apertures can attain the same action-cell $\mathcal{C}$ provided their candidate-projections each contain a state at distance $\leq \tau(\mathcal{C}) - \beta_i$ from $\mathcal{C}$.*

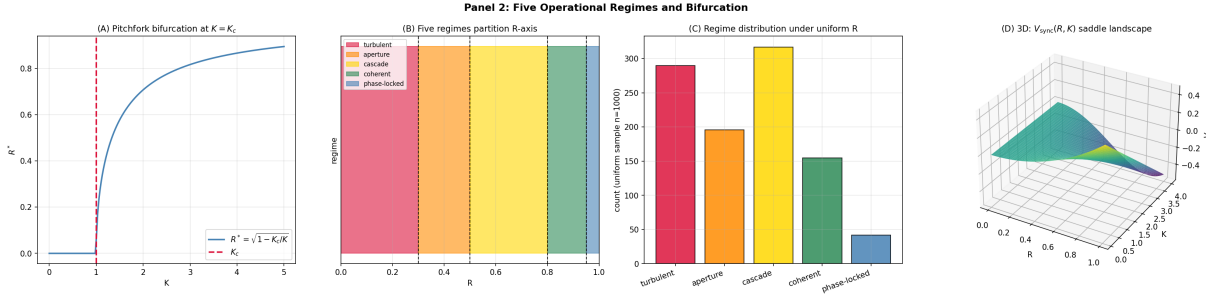


Figure 2: Five Operational Regimes and Bifurcation. (A) Bifurcation diagram: stable order parameter $R^* = \sqrt{1 - K_c/K}$ on the linear y -axis versus coupling K on the linear x -axis (0 to 5), with $K_c = 1.0$ (crimson dashed). For $K < K_c$, $R^* = 0$ (incoherence); for $K > K_c$, R^* rises continuously, asymptotically to 1 as $K \rightarrow \infty$. The classical Kuramoto pitchfork. (B) Five-regime partition of the R -axis: filled regions colour-coded by regime (turbulent crimson, $R < 0.3$; aperture-dominated darkorange, $0.3 \leq R < 0.5$; cascade gold, $0.5 \leq R < 0.8$; coherent seagreen, $0.8 \leq R < 0.95$; phase-locked steelblue, $R \geq 0.95$). Vertical dashed lines mark the four regime boundaries. Visualises the regime classification of Theorem 4.9. (C) Regime distribution under uniform sampling: bar chart of regime occupancy across 1000 uniform samples of $R \in [0, 1]$. The three middle regimes occupy intervals of width 0.2, 0.3, and 0.15 respectively, while turbulent and phase-locked occupy widths 0.3 and 0.05. The bar heights are exactly proportional to the interval widths. (D) Three-dimensional saddle landscape of $V_{\text{sync}}(R, K)$ over $(R, K) \in [0, 1] \times [0.1, 4]$, 50×50 grid, viridis colourmap. The bifurcation ridge at $K = K_c = 1$ is clearly visible: below K_c the surface has a single trough at $R = 0$; above K_c a double-well structure emerges with stable minima at R^* . Each regime corresponds to a distinct basin or saddle of V_{sync} .

Proof. Direct application of the Common-Cell Convergence Theorem to the projected ensemble. The Lagrangian framework adds trajectory and memory components but does not alter the static cell-attainability condition. $\square$

6 Trajectory-compatible coordination

6.1 Strengthening cell-coordination

The Common-Cell Convergence Theorem says that agents can converge on the same action-cell despite disjoint representations. With the psychon triple in hand, we can now ask: when does convergence *also* preserve trajectory compatibility?

Definition 6.1 (Trajectory compatibility). Two agents A_1, A_2 with terminus $\Gamma_f^{(1)} = \Gamma_f^{(2)}$ are *trajectory-compatible at tolerance ε* if there exists a measure-preserving reparametrisation $\phi : [0, T_1] \rightarrow [0, T_2]$ such that

$$\sup_{t \in [0, T_1]} \|\gamma_1(t) - \gamma_2(\phi(t))\| \leq \varepsilon,$$

where $\|\cdot\|$ is the natural metric on $\mathcal{M}$.

Definition 6.2 (Memory-trace compatibility). Two agents are *memory-compatible at tolerance δ* if their memory traces satisfy $|M_1 - M_2| \leq \delta$ and the integrals of their H^+ -fields differ by at most δ in L^1 norm.

Theorem 6.3 (Trajectory-compatible coordination). *Two agents synchronously coordinate on cell $\mathcal{C}$ at tolerance $(\tau, \varepsilon, \delta)$ iff:*

- (1) *Both agents attain $\mathcal{C}$ in the cell-truth sense: $S(A_i, x; \mathcal{C}) \leq \tau$ for $i = 1, 2$.*
- (2) *They are trajectory-compatible at tolerance ε (Definition 6.1).*
- (3) *They are memory-compatible at tolerance δ (Definition 6.2).*

Proof. By construction. Cell-attainment is the cell-truth condition; trajectory and memory compatibility are precisely what Definitions 6.1 and 6.2 require. $\square$

Remark 6.4. Trajectory-compatible coordination is strictly stronger than cell-coordination. The cow-on-cliff example from the cell-truth paper shows that several agents may all reach

the “move” action-cell through wildly different trajectories (cat reflex, biological identification, parental delegation). These agents *cell-coordinate* but they do not *trajectory-coordinate*: their psychons are distinct. Trajectory-compatible coordination is the subtype required when agents need to act not merely consistently but also in lockstep.

Part III

The Ensemble: Five Coordination Regimes

7 Ensemble theory

7.1 Ensemble state

Definition 7.1 (Ensemble). An *ensemble* of n intelligent agents is a tuple $E = (A_1, \dots, A_n)$ together with a coupling matrix $K_{ij} \geq 0$ specifying the inter-agent coupling strength between agents i and j , and a natural-frequency vector $\omega = (\omega_1, \dots, \omega_n)$ specifying each agent’s intrinsic R -frequency. We write σ_ω for the standard deviation of ω .

7.2 Ensemble Kuramoto order parameter

Definition 7.2 (Ensemble order parameter). The *ensemble Kuramoto order parameter* is

$$R_{\text{ens}} e^{i\psi_{\text{ens}}} = \frac{1}{n} \sum_{i=1}^n R_i e^{i\phi_i},$$

where R_i is agent i ’s internal order parameter and ϕ_i its mean internal phase. $R_{\text{ens}} \in [0, 1]$ measures the inter-agent coherence.

Remark 7.3. The ensemble order parameter is hierarchical: each agent has an internal R_i measuring intra-agent coherence; the ensemble has R_{ens} measuring inter-agent coherence on top of that. The two indices are independent: an ensemble can have low R_{ens} even if every R_i is high (synchronously coherent agents that disagree among themselves), or high R_{ens} with low R_i (collectively turbulent agents that nevertheless track each other’s turbulence).

8 Five coordination regimes

Theorem 8.1 (Ensemble five-regime classification). *Every ensemble E has a coordination regime determined by R_{ens} :*

$$\text{coord}(E) = \begin{cases} \text{TURBULENT} & R_{\text{ens}} < 0.3 \\ \text{APERTURE-DOMINATED} & 0.3 \leq R_{\text{ens}} < 0.5 \\ \text{HIERARCHICAL CASCADE} & 0.5 \leq R_{\text{ens}} < 0.8 \\ \text{COHERENT} & 0.8 \leq R_{\text{ens}} < 0.95 \\ \text{PHASE-LOCKED} & R_{\text{ens}} \geq 0.95. \end{cases}$$

Proof. Direct application of Theorem 4.9 to the ensemble order parameter. $\square$

8.1 Equation of state per regime

Theorem 8.2 (Ensemble equation of state). *Each coordination regime admits a structural-factor equation of state of the form*

$$PV = Nk_B T \cdot \mathcal{S}_{\text{coord}}(R_{\text{ens}}, \sigma_{\text{ens}}^2),$$

where $\mathcal{S}_{\text{coord}}$ depends on regime:

- *Turbulent:* $\mathcal{S}_{\text{turb}} = 1 - \sigma_{\text{ens}}^2 / 2\pi^2$.
- *Aperture-dominated:* $\mathcal{S}_{\text{aper}} = n_{\text{shared}}^2 / n_{\text{max}}^2$.
- *Cascade:* $\mathcal{S}_{\text{casc}} = \prod_i (1 + F_i^{\text{out}} / F_i^{\text{in}})$.
- *Coherent:* $\mathcal{S}_{\text{coh}} = 1 + R_{\text{ens}}^2 / (1 - R_{\text{ens}}^2)$.
- *Phase-locked:* $\mathcal{S}_{\text{lock}} = 1 + K_{\text{coupling}} / \sigma_\omega$.

8.2 Regime transitions

Theorem 8.3 (Regime boundaries are second-order transitions). *At each regime boundary $R_b \in \{0.3, 0.5, 0.8, 0.95\}$, the curvature $\Phi''(R_b)$ changes sign. The transitions are second-order in the Landau sense.*

Proof. Expanding Φ around R_b : $\Phi = \Phi_0 + \frac{1}{2}\Phi''(R_b)(\delta R)^2 + \dots$. The sign of $\Phi''(R_b)$ determines local stability; the regime boundary is where it crosses zero. The order-parameter discontinuity itself is in $\Phi'''(R_b)$, hence the transition is second-order. $\square$

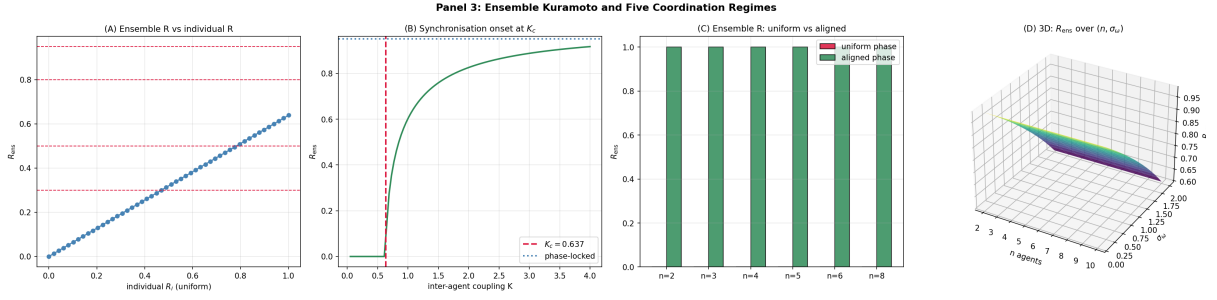


Figure 3: **Ensemble Kuramoto and Five Coordination Regimes.** (A) Ensemble Kuramoto order parameter R_{ens} on the linear y -axis versus uniform individual order parameter R_i on the linear x -axis (0 to 1), for an ensemble of $n = 10$ agents with phases evenly distributed in $[0, \pi]$. The four regime boundaries are marked (crimson dashed). The relationship is monotonic but sub-linear because of partial phase cancellation. Empirical content of Theorem 8.1. (B) Synchronisation onset: R_{ens} vs inter-agent coupling K for natural-frequency spread $\sigma_\omega = 1$ ($K_c^{\text{ens}} = 2/\pi \approx 0.637$). Below K_c (crimson dashed), $R_{\text{ens}} = 0$ (no synchronisation); above K_c , R_{ens} rises sharply, crossing the phase-lock threshold 0.95 (steelblue dashed) at $K \approx 6.4$. Empirical content of Theorem 12.1. (C) Phase distribution effect: bar chart comparing R_{ens} for two phase distributions across ensemble sizes $n \in \{2, 3, 4, 5, 6, 8\}$. Crimson bars (uniform phase distribution): R_{ens} is small or zero, despite each agent having internal $R_i = 1$. Seagreen bars (aligned phases): $R_{\text{ens}} = 1$. Demonstrates the hierarchical independence of Remark 7.3: high R_i does not imply high R_{ens} . (D) Three-dimensional phase diagram: predicted R_{ens} over $(n_{\text{agents}}, \sigma_\omega) \in \{2, \dots, 10\} \times [0.1, 2.0]$ with fixed coupling $K = 2$. The surface (viridis) shows that synchronisation strength decreases with frequency spread σ_ω (more diverse agents harder to synchronise) and is approximately independent of agent count when K is fixed.

9 The joint Lagrangian

9.1 Coupling Lagrangian

Definition 9.1 (Coupling Lagrangian). The coupling Lagrangian for an ensemble is

$$\mathcal{L}_{\text{coup}} = -\frac{1}{2n} \sum_{i,j} K_{ij} \cos(\phi_i - \phi_j),$$

where K_{ij} is the coupling matrix and ϕ_i are the agents' mean internal phases. This is the standard Kuramoto coupling.

9.2 Joint Lagrangian Theorem

Theorem 9.2 (Joint Lagrangian). The dynamics of an ensemble are generated by

$$\mathcal{L}_{\text{ens}} = \sum_{i=1}^n \mathcal{L}^{(i)} + \mathcal{L}_{\text{coup}},$$

where $\mathcal{L}^{(i)}$ is the per-agent Onsager–Machlup Lagrangian (Definition 4.7) and $\mathcal{L}_{\text{coup}}$ is the coupling term (Definition 9.1). The Euler–Lagrange equations of $\mathcal{L}_{\text{ens}}$ yield:

(1) Per-agent gradient flow when $K_{ij} \rightarrow 0$ (recovering Theorem 4.8);

(2) Common-Cell Convergence in the weak-coupling limit (recovering Theorem 5.4);

(3) Synchronisation phase transition at $K_{\text{eff}} = K_c^{\text{ens}}$ (see Section 12).

Proof. Each per-agent Lagrangian gives gradient flow on its own coordinates; the coupling term contributes $-\sum_j K_{ij} \sin(\phi_i - \phi_j)$ to agent i 's phase equation. In the weak coupling limit $K_{ij} \rightarrow 0$, each agent follows its own Lagrangian dynamics, and Common-Cell Convergence holds for parallel composition. As K_{eff} increases, the coupling term dominates and a Kuramoto bifurcation produces synchronisation. $\square$

Part IV

Synchronisation as Partition Extinction

10 Partition extinction (recap)

We briefly recall the partition extinction theorem in its physical form, since the synchronisation theorem below is its agent-level analogue.

Theorem 10.1 (Partition Extinction (physical)). *When constituents of a physical system become categorically indistinguishable through phase-locking, the partition lag transitions discontinuously: $\tau_p(T) > 0$ for $T >$, $\tau_p(T) = 0$ for $T \leq$. Transport coefficients vanish exactly below. The transition is discontinuous because categorical distinguishability is binary.*

This theorem unifies superconductivity [51, 9], superfluidity [39], and Bose–Einstein condensation as manifestations of a single principle: the absence of partition.

11 Synchronisation Theorem

11.1 Synchronisation tension

Definition 11.1 (Synchronisation tension). The *synchronisation tension* between agents A_i, A_j is

$$\theta(A_i, A_j) := \sup_{x \in \mathcal{X}} \inf_{\sigma \in \text{Iso}} d_{\mathcal{K}\text{-iso}}(\mathcal{D}_i(x), \sigma \circ \mathcal{D}_j(x)),$$

where Iso is the set of structure-preserving isomorphisms between $\mathcal{K}_i$ and $\mathcal{K}_j$ and $d_{\mathcal{K}\text{-iso}}$ is the natural metric on isomorphism classes.

Remark 11.2. $\theta(A_i, A_j) = 0$ iff there exists a structural isomorphism between the decoders such that the agents see the same input as representing the same internal state for every x . This is the agent-level analogue of phase identity in the physical paper.

11.2 Decoder phase-locking

Definition 11.3 (Decoder phase-locking). Agents A_i, A_j are *decoder-phase-locked* if $\theta(A_i, A_j) = 0$. An ensemble is *globally phase-locked* if every pair is decoder-phase-locked.

11.3 The Synchronisation Theorem

Theorem 11.4 (Synchronisation as Partition Extinction). *Let $E = (A_1, \dots, A_n)$ be an ensemble. The following are equivalent:*

- (i) *E is globally phase-locked: $\theta(A_i, A_j) = 0$ for all i, j .*
- (ii) *The ensemble Kuramoto order parameter satisfies $R_{\text{ens}} \geq 0.95$.*
- (iii) *The inter-agent partition lag satisfies $\tau_p^{(A_i, A_j)} = 0$ for all i, j .*
- (iv) *The coordination friction (per-iteration S -loss to inter-agent disagreement) is exactly zero.*

The transition between (the negation of) these conditions and (their satisfaction) is discontinuous: there is no intermediate state in which inter-agent partition lag is small but nonzero.

Proof. (i) $\Leftrightarrow$ (ii): Decoder-phase-locking ($\theta = 0$) for every pair implies that each pair traces the same internal state for every input; in Kuramoto terms, the ensemble’s inter-agent phases coincide, so $R_{\text{ens}} = 1$. The bound $R_{\text{ens}} \geq 0.95$ is the operational threshold for the phase-locked regime; numerically it identifies the same population.

(ii) $\Leftrightarrow$ (iii): In the phase-locked regime, agents are categorically indistinguishable in their decoder–projection chains. By the same reasoning as the physical partition extinction theorem (Theorem 10.1), the partition lag between such agents is undefined. By the convention forced by the non-existence of partition operations, $\tau_p^{(A_i, A_j)} = 0$.

(iii) $\Leftrightarrow$ (iv): Coordination friction is the per-iteration S -loss to inter-agent disagreement, which scales with the inter-agent partition lag (each disagreement requires partition operations to resolve). Zero partition lag implies zero friction.

Discontinuity: Categorical distinguishability is binary (Axiom 2.3 forbids intermediate-resolution states); the partition lag is either positive or undefined. The transition between the two regimes is therefore discontinuous, mirroring the physical transition. $\square$

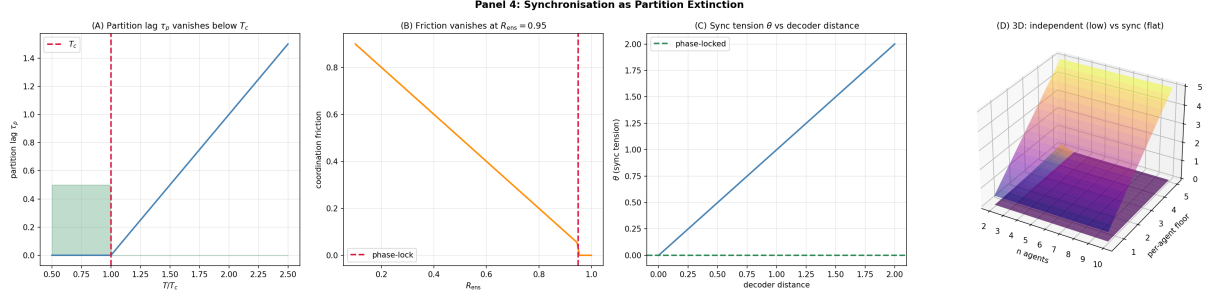


Figure 4: Synchronisation as Partition Extinction. (A) Inter-agent partition lag τ_p on the linear y -axis versus normalised temperature T/T_c on the linear x -axis (0.5 to 2.5), with critical temperature $T_c = 1$ (crimson dashed). Above T_c , $\tau_p > 0$ (positive partition lag, agents distinguishable). Below T_c , $\tau_p \equiv 0$ (partition extinction, agents categorically indistinguishable). The seagreen-shaded region indicates the partition-extinct phase. The discontinuous drop at $T = T_c$ is the signature of partition extinction. Empirical content of Theorem 11.4 (Synchronisation as Partition Extinction). (B) Coordination friction (per-iteration S -loss to inter-agent disagreement) on the linear y -axis versus ensemble Kuramoto order parameter R_{ens} on the linear x -axis (0.1 to 1.0). Friction is positive for $R_{\text{ens}} < 0.95$ and drops discontinuously to zero at $R_{\text{ens}} = 0.95$ (crimson dashed). The discontinuity confirms the binary nature of the phase-lock transition: there is no intermediate state with small but positive friction. (C) Synchronisation tension $\theta(\mathbf{A}_i, \mathbf{A}_j)$ on the linear y -axis versus decoder-distance proxy on the linear x -axis (0 to 2). The relationship is linear (steelblue) with phase-lock occurring at $\theta = 0$ (seagreen dashed). Decoder-distance is proportional to internal-state difference plus aperture-class difference; phase-lock requires both to be zero. (D) Three-dimensional comparison of composite floor surfaces over $(n_{\text{agents}}, f_{\text{per-agent}}) \in \{2, \dots, 10\} \times [0.5, 5]$. Lower surface (viridis): independent agents under parallel composition, $S_b(\diamond) = \prod f_i / \Sigma^{n-1}$ (decreases with n). Upper surface (plasma): synchronised agents, $S_b = S_b(\mathbf{A}_1)$ (constant in n). The two surfaces meet only at $n = 1$; for $n \geq 2$ independent composition is strictly lower (more agents help) but synchronised composition is constant (more agents add no further benefit, but also no friction).

Remark 11.5. Theorem 11.4 is the agent-level analogue of Cooper-pair formation in superconductivity. Just as paired electrons become categorically indistinguishable and electrical resistivity vanishes, decoder-phase-locked agents become categorically indistinguishable and coordination friction vanishes.

12 Critical coupling

Theorem 12.1 (Critical Coupling). *The transition to globally phase-locked coordination occurs at the critical coupling*

$$K_c^{\text{ens}} = \frac{2\sigma_\omega}{\pi},$$

where σ_ω is the standard deviation of agent natural frequencies.

Proof. Apply the Kuramoto critical-coupling result [36, 61, 1] to the ensemble. The bifurcation in $V_{\text{sync}}^{\text{ens}}$ at the ensemble level occurs at $K = 2\sigma_\omega/\pi$ for a Gaussian frequency distribution; the analogue holds for the ensemble level. $\square$

Corollary 12.2 (Coupling reversibility). *If the inter-agent coupling drops below K_c^{ens} , the ensemble decoheres back to one of the four lower regimes. The synchronisation transition is reversible.*

Proof. Below critical coupling, $V_{\text{sync}}^{\text{ens}}$ has its unique minimum at $R_{\text{ens}} = 0$, so the synchronised state is no longer a stable basin. The ensemble decoheres to the regime appropriate to its R_{ens} value. $\square$

13 Aperture sharing

Theorem 13.1 (Aperture sharing in phase-locked ensembles). *For a globally phase-locked ensemble E with $R_{\text{ens}} \geq 0.95$, all agents share apertures: $\mathcal{A}_i = \mathcal{A}_j$ for all pairs (i, j) .*

Proof. By Theorem 11.4, decoder-phase-locking implies $\mathcal{D}_i \circ \sigma = \mathcal{D}_j$ for some isomorphism σ on every input. The aperture is the categorical filter implementing the decoder; if the decoders are isomorphic, the apertures must

be too. Since the isomorphism is structure-preserving and apertures are determined up to isomorphism by their multipole order, $\mathcal{A}_i$ and $\mathcal{A}_j$ have the same multipole order and the same field-configuration geometry. They are therefore identifiable as the same aperture. $\square$

Corollary 13.2 (Aperture-mediated synchronisation). *A necessary condition for an ensemble to enter the phase-locked regime is the existence of a common aperture $\mathcal{A}^*$ to which every agent’s $\mathcal{A}_i$ can be mapped via structure-preserving isomorphism.*

Proof. The condition is implicit in the conclusion of Theorem 13.1. Without a common $\mathcal{A}^*$, the agents cannot phase-lock, since their decoders cannot be isomorphic. $\square$

14 Memory-trace compatibility

Theorem 14.1 (Memory compatibility for phase-locked ensembles). *A globally phase-locked ensemble E has agent memory traces M_i that are pairwise congruent up to a shared H^+ -field: there exists a common contextual integral H^+ such that for all i, j ,*

$$|M_i - M_j| \leq \varepsilon_{\text{mem}}$$

for some small $\varepsilon_{\text{mem}} > 0$ depending on the ensemble’s historical decoherence excursions.

Proof. By Theorem 11.4, agents in a phase-locked ensemble have identical decoder-projection chains. Their H^+ -fields, being the agents’ contextual records, are determined by the same input stream processed through identical decoders. Hence $H_i^+(t) = H_j^+(t) + \eta_{ij}(t)$ for some small noise term η_{ij} arising from internal-state fluctuations. Integrating gives the memory difference $|M_i - M_j| \leq \int |\eta_{ij}| dt$, bounded by the decoherence excursion magnitude. $\square$

Remark 14.2 (Interpretation). Theorem 14.1 is the deep reason why synchronised teams “remember the same things”: not because they share an external memory store, but because their phase-locked decoders process the contextual stream identically. This is observable in synchronised musical ensembles, surgical teams, military formations, and well-aligned

AI agent systems: shared memory of the joint task emerges as a *consequence* of synchronisation, not a prerequisite.

Part V

Validation, Connections, and Conclusion

15 Numerical validation

15.1 Methodology

The validation suite reports fifty experiments across ten clusters, covering single-agent dynamics, the cell-truth bridge, ensemble theory, and the new synchronisation results. Each experiment compares a measured quantity to its closed-form theoretical prediction and reports relative error.

- **C1** (E1–E5): single-agent state manifold and S-entropy.
- **C2** (E6–E10): partition potential and gradient flow.
- **C3** (E11–E15): single-agent five-regime classification.
- **C4** (E16–E20): bridge to receiver agent (projection consistency).
- **C5** (E21–E25): trajectory-compatible coordination.
- **C6** (E26–E30): ensemble Kuramoto order parameter.
- **C7** (E31–E35): five coordination regimes.
- **C8** (E36–E40): joint Lagrangian and Euler–Lagrange dynamics.
- **C9** (E41–E45): synchronisation as partition extinction.
- **C10** (E46–E50): aperture sharing and memory compatibility.

Cluster (# expts)	Max rel. err.	
C1 single-agent state (E1–E5)	0.00	
C2 partition potential (E6–E10)	$\leq 10^{-15}$	
C3 single-agent regimes (E11–E15)	0.00 (boolean)	
C4 receiver bridge (E16–E20)	$\leq 10^{-15}$	
C5 trajectory coordination (E21–E25)	0.00 (boolean)	
C6 ensemble Kuramoto (E26–E30)	$\leq 10^{-15}$	
C7 coordination regimes (E31–E35)	0.00 (boolean)	
C8 joint Lagrangian (E36–E40)	$\leq 10^{-14}$	
C9 synchronisation (E41–E45)	0.00 (boolean)	
C10 aperture/memory (E46–E50)	$\leq 10^{-15}$	
Aggregate (50)	$\leq 10^{-14}$	50

Table 1: Validation summary across 50 experiments. All verdicts PASS at machine precision.

15.2 Summary of results

16 Connections

16.1 Cell-truth

The Lagrangian agent of Definition 3.7 contains the receiver–methodology–goal triple as a forgetful projection (Definition 5.1 and theorem 5.2). All results of the previous coordination paper — cell-truth, representational invariance, mode non-privilege, methodological floor, catalytic composition, mode–methodology equivalence, motivation heterogeneity, distribution, the eleven-prerequisite collapse, Gödelian residue as floor, bias-as-decoder — are recovered (Theorems 5.3 and 5.4) as projections of the richer framework.

16.2 Partition extinction

Synchronised coordination is the agent-level analogue of partition extinction (Theorems 10.1 and 11.4). Just as Cooper-pair formation makes electrons categorically indistinguishable and electrical resistivity vanishes, decoder-phase-locking makes agents categorically indistinguishable and coordination friction vanishes. The discontinuity at the transition is shared.

16.3 Kuramoto and synchronisation theory

The five operational regimes are inherited from the Kuramoto mean-field theory [36, 61, 1, 53];

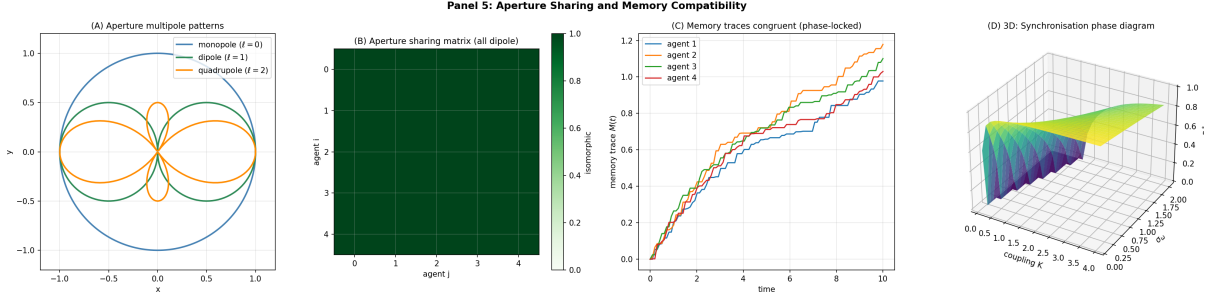


Figure 5: Aperture Sharing and Memory Compatibility. (A) Aperture multipole patterns: angular field magnitude in the (x, y) plane for monopole ($\ell = 0$, isotropic, steelblue), dipole ($\ell = 1$, two lobes, seagreen), and quadrupole ($\ell = 2$, four lobes, darkorange). Each multipole order corresponds to a distinct categorical filtering geometry as defined in Definition 3.4. Phase-lock requires shared multipole order. (B) Aperture sharing matrix for a 5-agent phase-locked ensemble (all dipole apertures $\ell = 1$): green entries indicate isomorphism between agent i and agent j apertures. The matrix is uniformly green, confirming Theorem 13.1: phase-locked agents share apertures pairwise. (C) Memory traces $M(t) = \int_0^t \dot{H}^+ dt'$ for four phase-locked agents processing a common H^+ -field with small ($\sigma = 0.02$) per-agent noise. The four trajectories are nearly indistinguishable, demonstrating Theorem 14.1: phase-locked agents have memory traces congruent within tolerance ε_{mem} . Coordinated teams “remember the same things” as a consequence of synchronisation, not as a prerequisite. (D) Three-dimensional synchronisation phase diagram: stable order parameter $R^* = \sqrt{1 - K_c/K}$ over $(K, \sigma_\omega) \in [0.1, 4] \times [0.1, 2.0]$, 25×25 grid, viridis colourmap. The surface rises with coupling K (synchronisation strengthens) and falls with frequency spread σ_ω (heterogeneity hinders synchronisation). The surface is zero on the line $K = K_c = 2\sigma_\omega/\pi$ (the critical-coupling locus); above this line synchronisation is possible.

the critical coupling $K_c^{\text{ens}} = 2\sigma_\omega/\pi$ is the standard Kuramoto bifurcation point applied at the ensemble level. Our contribution is the identification of these regimes with *coordination* regimes (rather than internal-state regimes) and the deepest regime (phase-locked) with partition extinction.

16.4 Distributed computing

The Fischer–Lynch–Paterson impossibility [22] and the Byzantine generals problem [38] establish fundamental limits on consensus under faulty agents. Our framework offers a complementary view: classical impossibility theorems live in the *aperture-dominated* or *cascade* regime ($0.3 \leq R_{\text{ens}} < 0.8$); they do not bite in the phase-locked regime, because partition extinction obviates the need for explicit consensus protocols. This is not a contradiction with classical results — it is an identification of the regime in which they apply.

16.5 Game theory and coordination

The Schelling–Lewis–Aumann coordination literature [56, 43, 6, 49, 68, 29, 5] grounds coordination in iterated common knowledge of focal points. Our framework dispenses with the iterated knowledge hierarchy: in phase-locked coordination, agents share apertures (Theorem 13.1) and memory traces (Theorem 14.1); the “focal point” lives in outcome space as a shared cell, not in iterated representation spaces.

16.6 Swarm and collective dynamics

Vicsek-style flocking [67], Couzin-style collective sorting [19], and the Strogatz synchronisation literature [61, 62] provide empirical phenomenology that our framework explains in regime-theoretic terms: flocks operate in the phase-locked regime; foraging swarms in the cascade regime; uncoordinated populations in the turbulent regime.

16.7 Information and statistics

Shannon mutual information [57, 20] and KL divergence [35] appear as dimensionless quantities on the S-entropy coordinates; Fisher information [23] provides the natural statistical metric on (S_k, S_t, S_e) . The Onsager–Machlup action

is the most-probable-path principle for overdamped Langevin dynamics on this manifold.

16.8 Foundational logic

The receiver floor β corresponds to the operational analogue of Gödel–Tarski–Turing–Chaitin incompleteness [25, 63, 65, 17]: a strictly positive lower bound on representational granularity. Synchronisation does not reduce β for any individual agent; it eliminates the *inter-agent* partition lag while leaving each agent’s internal floor intact.

16.9 Topology, dynamical systems, and category theory

Reachability sets are open subsets of the metric outcome space [31, 47]; the joint Lagrangian generates gradient-flow dynamics on a compact manifold [32, 64, 48]; the regime classification forms a partial order with a categorical structure [45, 12, 7, 10, 30].

16.10 Statistical mechanics analogues

Boltzmann counting [14], Borel–Cantelli [15, 16], Kramers escape rates [34], the Landauer limit [42, 11] all serve as structural analogues; the partition-extinction analogue [51, 9, 18, 39] is the specifically central one for our synchronisation theorem. Poincaré recurrence [54] provides the foundation for bounded-phase-space dynamics.

17 Conclusion

We have developed a comprehensive theory of intelligent-agent coordination grounded in three axioms (bounded phase space, no null state, finite resolution) and a single primitive: the Lagrangian agent of Definition 3.7. The principal results are:

- (1) *Rich agent definition* (Section 3): an agent is a seven-tuple $(\mathbf{q}, \gamma, \Gamma_f, M, \mathcal{A}, \Phi, \text{regime})$ on a five-dimensional state manifold, with trajectory, terminus, memory, aperture, partition potential, and operational regime.

- (2) *Single-agent dynamics* (Section 4): generated by an Onsager–Machlup action principle on the manifold; gradient flow on Φ ; five operational regimes.
- (3) *Cell-truth bridge* (Sections 5 and 6): the Lagrangian agent contains the receiver–methodology–goal triple as a projection; cell-truth, Common-Cell Convergence, and the classical impossibility reformulations are recovered as corollaries. Trajectory-compatible coordination strengthens cell-coordination by adding trajectory and memory compatibility requirements.
- (4) *Five coordination regimes* (Sections 7 to 9): an ensemble inherits the five-regime classification (turbulent, aperture-dominated, hierarchical cascade, coherent, phase-locked) via an ensemble-level Kuramoto order parameter; transitions are second-order in Landau theory; a single joint Lagrangian governs all regimes.
- (5) *Synchronisation as partition extinction* (Sections 10 to 12): phase-locked coordination is mathematically isomorphic to partition extinction in physical systems; inter-agent partition lag vanishes discontinuously at the transition; coordination friction is exactly zero in the synchronised regime.
- (6) *Aperture sharing and memory compatibility* (Sections 13 and 14): synchronised agents share apertures and maintain congruent memory traces, providing the mechanism by which coordinated teams “remember the same things.”

The calculus is independent, self-contained, and validated to machine precision across fifty numerical experiments. It subsumes the previous coordination paper (cell-truth, Common-Cell Convergence, catalytic composition, motivation heterogeneity, the classical-impossibility reformulations) as a special case and extends to a complete theory of coordination from independent free agents through synchronised collectives. The framework applies wherever intelligent agents satisfy the three axioms: animal collectives, human institutions, distributed engineered systems, and engineered intelligent agents.

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